

# Exact Phase-Type Point Processes: Certified Measurement of News Absorption and State-of-the-Art Prediction of Market Event Streams

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Paper under double-blind review

## Abstract

Temporal point processes are used both to *measure* mechanisms—how fast does a market absorb news, and how much of the reaction is the market reacting to itself?—and to *predict* the next event. We show that a single algebraic object, the phase-type (Erlang-chain) state space, serves both purposes with exact likelihoods. For measurement, we build a multiscale phase-type Hawkes process whose parameters all enter linearly; the log-likelihood is concave, the global maximum comes with a Fenchel duality-gap certificate, and a Markov embedding gives the echo-corrected response to an exogenous shock in closed form, stable if and only if the spectral radius of the branching matrix is below one. On six markets around 670 scheduled macroeconomic releases (2022–2026), with a built-in placebo test, the direct reaction to CPI, payrolls, PPI and the FOMC is absorbed in about a second, whereas echoes stretch half of the response to 16–137 s. For prediction, we introduce EPT-X, a neural point process that superposes a history-conditioned, defective hyper-Erlang renewal channel on a linear Hawkes backbone; its intensity and compensator remain in closed form, so it is trained and evaluated on the exact likelihood. EPT-X attains the best test likelihood on all five LOBSTER limit-order-book streams against six classical estimators and seven neural baselines, improves the best published result on Retweet by more than 0.6 nats per event, and matches it on Taobao and StackOverflow. We also find that a widely used IntensityFree implementation clamps inter-event times before scoring, which overstates its likelihood on streams with many short gaps; scored on the observed gaps, its order-book ranking changes.

## 1 Introduction

Event streams in markets—price changes, order-book messages, trades—are modelled with temporal point processes for two different purposes. Econometricians use Hawkes processes (Hawkes, 1971) to *measure* mechanisms: the branching structure separates exogenous, news-driven events from endogenous ones triggered by earlier events, and the fitted kernels quantify how fast information is absorbed (Hardiman et al., 2013; Filimonov & Sornette, 2012; Rambaldi et al., 2015). Machine-learning work uses neural temporal point processes to *predict* the next event and is judged by held-out likelihood on shared benchmarks (Du et al., 2016; Mei & Eisner, 2017; Zuo et al., 2020; Shchur et al., 2020; Xue et al., 2024; Chang et al., 2025). The two communities rarely meet, and each inherits a weakness. Measurement relies on non-convex fits whose optimality is unknown and on summaries—such as the “half-life”  $\ln 2/\beta$  of a news kernel—that ignore the echoes the news triggers. Prediction relies on neural intensities whose compensator  $\int \lambda$  is estimated by Monte Carlo, so that reported likelihoods carry sampling error and, as we show, are sensitive to implementation details.

We argue that one algebraic object serves both purposes exactly: the *phase-type state space*, in which every kernel is a nonnegative combination of Erlang densities on a logarithmic grid of time scales. Between events such a state evolves by an explicit matrix exponential, so intensities and compensators are available in closed form, and the class is dense in the densities on  $\mathbb{R}_+$ .

## Contributions.

1. **Certified measurement (MSX).** A multiscale phase-type Hawkes process with exogenous, marked news kernels in which all parameters enter linearly. The log-likelihood is concave; we compute the global maximum with accelerated EM and a barrier-Newton polish and return a Fenchel duality gap that certifies optimality (Proposition 2).
2. **Echo-corrected responses in closed form.** A Markov embedding turns the mean response to a shock into  $Ce^{At}z_0$ ; it is stable if and only if  $\rho(G) < 1$  for the branching matrix  $G$  (Theorem 3), and yields absorption quantiles, echo amplification and price-discovery curves without simulation.
3. **Exact neural prediction (EPT-X).** A neural point process whose latent state is phase-type, with a linear Hawkes backbone and a history-conditioned *defective hyper-Erlang renewal channel* superposed as a competing risk. The compensator stays exact, the model nests the Hawkes backbone, and the renewal channel is universal for the conditional gap law (Proposition 5).
4. **Benchmarks and a benchmark correction.** EPT-X is best on all five LOBSTER order-book streams and on Retweet, and statistically tied with the best published model on Taobao and Stack-Overflow. We find that a widely used IntensityFree implementation clamps inter-event times at  $10^{-5}$  s before scoring; re-scored on the observed gaps, its lead on two order-book streams disappears.
5. **Application.** On six markets around 670 scheduled releases (2022–2026), with a placebo test built into the likelihood, we find that the direct reaction to CPI, payrolls, PPI and the FOMC is absorbed in about a second while echoes stretch half of the response to 16–137 s and ninety percent to 5–11 minutes.

Linear-in-parameters kernel dictionaries (Bacry et al., 2015), EM for Hawkes processes (Lewis & Mohler, 2011; Zhou et al., 2013) and Erlang-chain embeddings are known individually; our contribution is their combination into certified, closed-form measurement and into an exact-likelihood neural model, together with the evidence that both work.

## 2 Related work

**Hawkes estimation.** Exponential and sum-of-exponential kernels are fitted by maximum likelihood (Bacry et al., 2015); nonparametric alternatives include EM on histogram kernels (Lewis & Mohler, 2011), ADM4 with low-rank and sparse penalties (Zhou et al., 2013), the conditional-law (Wiener–Hopf) method (Bacry et al., 2016), cumulant matching (Achab et al., 2018) and INAR approximations (Kirchner, 2017). None of these returns a certificate of global optimality for multiscale kernels with humps, and seasonality mis-specification is known to bias branching ratios upward (Filimonov & Sornette, 2015; Omi et al., 2017).

**News in Hawkes models.** Rambaldi et al. (2015) add an exogenous kernel for macroeconomic announcements to an FX Hawkes model; Andersen et al. (2003) document the price impact of announcement surprises. Absorption speed is typically read off the exogenous kernel alone. We give the echo-corrected response in closed form and test the news kernels against matched placebo windows inside the likelihood.

**Neural temporal point processes.** RMTTP (Du et al., 2016), NHP (Mei & Eisner, 2017), SAHP (Zhang et al., 2020), THP (Zuo et al., 2020), AttNHP (Yang et al., 2022) and the state-space S2P2 (Chang et al., 2025) parameterise the intensity with recurrent, attention or state-space encoders and estimate the compensator by Monte Carlo or quadrature; IntensityFree (Shchur et al., 2020) instead models the next-gap density with a log-normal mixture. Recent work explores Mamba-style encoders (Gao et al., 2024) and interpretable latent Hawkes dynamics (Boyd et al., 2025). EasyTPP (Xue et al., 2024) provides shared implementations, and Bosser & Ben Taieb (2023) study the predictive accuracy of neural TPPs. EPT-X differs in that its intensity between events is a read-out of an explicitly solvable phase-type system with bounded closed-form hazard terms, plus a renewal channel whose survival function is a closed-form mixture of Erlang survivals—so the likelihood we train and report is exact—and in that its backbone is the linear Hawkes process itself.

**Phase-type distributions.** Mixtures of Erlang distributions are dense in the distributions on  $(0, \infty)$  (Tijms, 1994; Lee & Lin, 2010); we use this fact for the renewal channel and the Erlang-chain recursion for all states.

### 3 Phase-type state spaces

**Dictionary.** Fix rates  $\beta_1, \dots, \beta_K$  on a logarithmic grid and Erlang orders  $r = 0, \dots, R-1$ . The normalised basis densities

$$g_{kr}(t) = \beta_k \frac{(\beta_k t)^r}{r!} e^{-\beta_k t}, \quad \int_0^\infty g_{kr} = 1, \quad (1)$$

generate kernels  $\phi(t) = \sum_{k,r} a_{kr} g_{kr}(t)$  with  $a_{kr} \geq 0$  and mass (branching ratio)  $\sum_{k,r} a_{kr}$ . Exponential mixtures ( $R = 1$ ) approximate every completely monotone kernel, including power laws (Bernstein’s theorem); higher orders add delayed humps ( $\phi(0) = 0$ ); and phase-type densities are dense in the densities on  $\mathbb{R}_+$ .

**Exact recursion.** All states used in this paper—Hawkes excitation, news response, the neural latent state of EPT and the renewal channel of EPT-X—are Erlang chains, for which evolution and integrals between events are closed form.

**Lemma 1** (Erlang-chain recursion). *Let  $s_r(t) = \sum_{t_n < t} g_{kr}(t - t_n)$  for one source and one rate  $\beta$ . Between events,  $\mathbf{s} = (s_0, \dots, s_{R-1})$  solves  $\dot{\mathbf{s}} = \beta(N - I)\mathbf{s}$  with  $N$  the down-shift, and at an event  $s_0 += \beta$ . Hence, with  $y = \beta\Delta$  and  $w_j = e^{-y}y^j/j!$ ,*

$$s_r(t + \Delta) = \sum_{j \leq r} w_j s_{r-j}(t), \quad \int_t^{t+\Delta} s_r = \frac{1}{\beta} \sum_{j \leq r} P(j+1, y) s_{r-j}(t),$$

where  $P$  is the regularised lower incomplete gamma function.

*Proof.*  $\frac{d}{dt}g_{kr} = \beta(g_{k,r-1} - g_{kr})$  for  $r \geq 1$  and  $\frac{d}{dt}g_{k0} = -\beta g_{k0}$ , with  $g_{kr}(0) = \beta\mathbb{I}[r = 0]$ . The matrix exponential of  $\beta(N - I)\Delta$  is  $e^{-y} \sum_{j < R} \frac{y^j}{j!} N^j$  because  $N$  is nilpotent and commutes with  $I$ ; integrating term by term gives  $P(j+1, y)/\beta$ .  $\square$

The cost of evaluating a likelihood is therefore  $O(NdKR^2)$  for  $N$  events in  $d$  dimensions, with no Monte-Carlo or quadrature error.

### 4 MSX: certified measurement with echo-corrected responses

**Model.** Consider  $d$  event dimensions observed in windows  $w$  with likelihood intervals  $[t_0^w, t_1^w]$ , and exogenous shocks  $e$  (here: scheduled releases) at times  $\tau_e$  with type  $c_e$  and nonnegative mark features  $\mathbf{z}_e = (1, z_e^+, z_e^-)$ , the standardised surprise split by sign. The MSX intensity of dimension  $i$  is

$$\begin{aligned} \lambda_i(t) = & \underbrace{c_{iw} + \sum_m \eta_{im} h_m(\text{tod}(t))}_{\text{baseline}} + \underbrace{\sum_j \sum_{t_n^j < t} \sum_{k,r} a_{ijk} g_{kr}(t - t_n^j)}_{\text{endogenous}} \\ & + \underbrace{\sum_{e: \tau_e < t} \sum_{k,r,m} b_{ic_ekrm} z_{em} g_{kr}^{\text{exo}}(t - \tau_e)}_{\text{exogenous (news)}} + \underbrace{\sum_{e: \tau_e > t} \sum_k \alpha_{ic_ek} \beta_k^{\text{ant}} e^{-\beta_k^{\text{ant}}(\tau_e - t)}}_{\text{anticipation}}, \end{aligned} \quad (2)$$

with per-window intercepts  $c_{iw}$ , periodic hat functions  $h_m$  of local time of day, and all parameters nonnegative. Writing  $\mathbf{x}(t) \geq 0$  for the stacked features,  $\lambda_i(t) = \mathbf{x}(t)^\top \boldsymbol{\theta}_i$ , and the log-likelihood separates across targets,

$$\ell_i(\boldsymbol{\theta}_i) = \sum_{n: u_n = i} \log(\mathbf{x}(t_n)^\top \boldsymbol{\theta}_i) - \boldsymbol{\Phi}^\top \boldsymbol{\theta}_i, \quad \boldsymbol{\Phi} = \sum_w \int_{t_0^w}^{t_1^w} \mathbf{x}(t) dt, \quad (3)$$

where every entry of  $\mathbf{x}(t_n)$  and  $\boldsymbol{\Phi}$  follows from Lemma 1.

**Concavity and a certificate.** Problem equation 3 maximises a concave function over the nonnegative orthant. Exposure-weighted  $\ell_1$  penalties are linear,  $\gamma^\top \theta$ , and are absorbed into  $\Phi$ . We run the multiplicative EM map  $\theta_p \leftarrow \theta_p (\sum_n x_{np}/\lambda_n)/(\Phi_p + \gamma_p)$ , accelerated by SQUAREM (Varadhan & Roland, 2008), and polish with a log-barrier Newton method.

**Proposition 2** (Duality-gap certificate). *For any feasible  $\theta \geq 0$  with intensities  $\lambda_n = \mathbf{x}_n^\top \theta > 0$ , let  $s = \max_p (X^\top \lambda^{-1})_p / (\Phi_p + \gamma_p)$ . Then*

$$\ell^* - \ell(\theta) \leq N \log s + (\theta^\top (\Phi + \gamma) - N),$$

and both terms vanish at the optimum.

The proof (Appendix A) applies Fenchel’s inequality  $\log u \leq vu - 1 - \log v$  to each  $\lambda_n$ . Every fit reported here carries this certificate.

**Markov embedding.** Stack the endogenous states into  $\mathbf{x}$ , let  $F$  be the block-diagonal Erlang-chain generator,  $G_{\text{in}}$  inject  $\beta_k$  into phase  $r = 0$  of a source, and  $\Theta$  the endogenous weights. The *mean* excess intensity caused by one shock with exogenous states  $\mathbf{y}$  ( $\dot{\mathbf{y}} = F_e \mathbf{y}$ ,  $\mathbf{y}(0) = \mathbf{y}_0$ ) solves

$$\dot{\mathbf{z}} = \mathcal{A} \mathbf{z}, \quad \mathcal{A} = \begin{pmatrix} F + G_{\text{in}} \Theta & G_{\text{in}} \Theta_e \\ 0 & F_e \end{pmatrix}, \quad r(t) = \begin{pmatrix} \Theta & \Theta_e \end{pmatrix} \mathbf{z}(t), \quad \mathbf{z}(0) = \begin{pmatrix} 0 \\ \mathbf{y}_0 \end{pmatrix}. \quad (4)$$

**Theorem 3** (Stability). *Let  $G_{ij} = \sum_{k,r} a_{ijk r}$  be the branching matrix. Then  $F + G_{\text{in}} \Theta$  is Hurwitz if and only if  $\rho(G) < 1$ .*

*Proof.*  $-F$  is a nonsingular M-matrix and  $G_{\text{in}} \Theta \geq 0$ , so  $F + G_{\text{in}} \Theta$  is Hurwitz iff  $\rho(-F^{-1} G_{\text{in}} \Theta) < 1$  (regular splitting; Varga, 2000, Thm. 3.29). Moreover  $\rho(-F^{-1} G_{\text{in}} \Theta) = \rho(\Theta(-F)^{-1} G_{\text{in}})$  and  $(\Theta(-F)^{-1} G_{\text{in}})_{ij} = \sum_{k,r} a_{ijk r} \int_0^\infty g_{kr} = G_{ij}$ .  $\square$

**Corollary 4** (Closed-form absorption). *If  $\rho(G) < 1$ , the echo-corrected response is  $r(t) = C e^{A t} \mathbf{z}_0$  with total  $-C A^{-1} \mathbf{z}_0$ ; the fraction absorbed by time  $t$  is  $F(t) = C A^{-1} (e^{A t} - I) \mathbf{z}_0 / (C A^{-1} (-\mathbf{z}_0))$ , its quantiles  $t_q = F^{-1}(q)$  are the echo-corrected absorption times, and the amplification is  $\int r / \int r_{\text{direct}}$  with  $r_{\text{direct}}(t) = \Theta_e e^{F_e t} \mathbf{y}_0$ . For a price observed through up and down  $\delta$ -crossings  $(u, v)$ , the expected price-discovery curve is  $\delta \int_0^t (r_u - r_v)$ .*

The usual “news half-life”  $\ln 2 / \beta$  describes only  $r_{\text{direct}}$ ; when  $\rho(G)$  is close to one the echoes dominate, and Section 7 shows that the difference is one to two orders of magnitude.

## 5 EPT-X: exact neural prediction

**EPT: a neural phase-type state.** The latent state  $\mathbf{x}(t) \in \mathbb{R}_{\geq 0}^P$  consists of  $C$  channels of Erlang- $R$  chains with learnable rates  $\beta_{ck}$ , evolving between events by Lemma 1. At event  $n$  with mark  $m_n$  and gap  $\Delta_n$ , a GRU summary  $\mathbf{h}_n$  of the history (inputs: a mark embedding and log transforms of the gap; EPT-X uses the gap’s standardised logarithm, which separates nanosecond ties from zero) produces a gate and a nonnegative jump,  $\mathbf{x}(t_n^+) = \sigma(W_g \mathbf{h}_n) \odot \mathbf{x}(t_n^-) + \text{softplus}(W_j \mathbf{h}_n)$ . The marked intensities are

$$\lambda_m^{\text{EPT}}(t) = \mu_m(\mathbf{h}_n) + \sum_p C_{mp}(\mathbf{h}_n) x_p(t) + g_{1m}(\mathbf{h}_n) e^{-a(\mathbf{h}_n)s} + g_{2m}(\mathbf{h}_n) (1 - e^{-\kappa(\mathbf{h}_n)s}) + [\Theta \mathbf{x}^H(t)]_m,$$

with  $s = t - t_n$ , all coefficients nonnegative, and  $\mathbf{x}^H$  the state of a *linear multivariate phase-type Hawkes process* on a fixed log grid of rates, which receives the jump  $\beta_k$  in the source’s first phase at every event. The last term is a residual path: with the neural read-outs at zero the model is the Hawkes process of Section 4, so the network only learns departures from linear excitation. Bounded hazard heads replace an unbounded Gompertz term, which overflows on heavy-tailed gaps.

**The renewal channel.** Order-book and social streams contain cascades: several events within a tiny interval, followed by a pause. A linear Hawkes intensity describes excitation *between* cascades but has no direct handle on the gap distribution *within* them. EPT-X adds, by superposition, a channel that models the next gap directly. Fix  $J$  Erlang atoms  $(R_j, \beta_j)$  whose means  $R_j/\beta_j$  lie on a logarithmic grid spanning the data’s time scales, with orders  $R_j \in \{1, 4, 16\}$  (optionally up to 1024; coefficient of variation  $R_j^{-1/2}$ ). From  $\mathbf{h}_n$  the network outputs weights  $\mathbf{w}$  on the simplex over the  $J$  atoms and a *defect* atom  $w_\infty$ , a mark law  $q_{m|j}$  per time scale and, optionally, a bounded log-rate shift that lets each atom slide within its grid cell. The channel’s joint density of the next gap  $s$  and mark  $m$  and its survival function are

$$p(s, m) = \sum_{j=1}^J w_j q_{m|j} \frac{\beta_j^{R_j} s^{R_j-1} e^{-\beta_j s}}{(R_j - 1)!}, \quad S_X(s) = w_\infty + \sum_{j=1}^J w_j e^{-\beta_j s} \sum_{i < R_j} \frac{(\beta_j s)^i}{i!},$$

and EPT-X has intensity  $\lambda_m(t) = \lambda_m^{\text{EPT}}(t) + p(s, m)/S_X(s)$ .

**Proposition 5.** (i) *The compensator of EPT-X over  $(t_n, t_n + \Delta]$  is  $\Lambda^{\text{EPT}}(\Delta) - \log S_X(\Delta)$ , in closed form.* (ii) *With  $w_\infty = 1$  EPT-X equals EPT, which nests the linear phase-type Hawkes process.* (iii) *Given  $\mathbf{h}_n$ , the renewal channel is the absorption time of a finite continuous-time Markov chain, so its survival, quantiles and moments are closed-form matrix functionals.* (iv) *Mixtures of Erlang distributions are dense in the distributions on  $(0, \infty)$  (Tijms, 1994; Lee & Lin, 2010), so the channel can approximate any conditional gap law as the dictionary is refined—the expressiveness of a conditional mixture density (Shchur et al., 2020), with an exact intensity and compensator.*

*Proof.* (i) The compensator of a superposition is the sum of compensators, and that of a renewal hazard  $p/S_X$  is  $-\log S_X$ . (ii)  $w_\infty = 1$  forces  $w_j = 0$ , hence  $p \equiv 0$  and  $S_X \equiv 1$ . (iii) An Erlang mixture with a defect is a phase-type law. (iv) is the cited density result applied to the mixture over atoms.  $\square$

**Exact training and Bayes-optimal prediction.** Training maximises the exact log-likelihood  $\sum_n \log \lambda_{m_n}(t_n) - \Lambda$ , so gradients carry no sampling noise; for large orders  $S_X$  is evaluated through the regularised upper incomplete gamma function, whose gradient we guard at zero lags. The log-likelihood splits exactly into a time term  $(\log \lambda(t_n) - \Lambda)$  and a mark term  $(\log \lambda_{m_n}/\lambda)$ , which we report separately. Next-event predictions are Bayes-optimal functionals of the exact density,  $\mathbb{E}[\Delta] = \int_0^\infty S$  and  $\hat{m} = \arg \max_m \int \lambda_m S$ , computed by one-dimensional quadrature on a logarithmic grid.

## 6 Experiments

All tables are generated from result files written by the experiment scripts; every configuration choice is made on validation data, and the test split is scored once.

### 6.1 Synthetic recovery

Five ground truths, simulated by exact cluster sampling, deliberately include kernels outside the MSX class: S1 a single exponential, S2 a power law (Lomax, tail exponent 1.5), S3 a gamma hump plus a fast exponential, S4 a sparse four-dimensional network with 10 ms and 30 s components, and S5 a sparse ten-dimensional power-law network; ten windows per split and five seeds. Competitors are the exponential and sum-of-exponential MLEs, the nonparametric EM, ADM4, the conditional law and NPHC. MSX-auto selects the Erlang order and  $\ell_1$  level on held-out training windows.

Table 1 reports the held-out likelihood deficit to the true model. When the truth is a single exponential, the correctly specified single-decay estimators win, as they must; under a power law the sum-of-exponential estimators are tied. Whenever the kernel has structure outside a single family—a delayed hump, two separated time scales or a power-law network—MSX-auto is best, with deficits 7–10 times smaller than the best competitor’s, a branching-matrix error roughly halved and perfect recovery of the causal graph (Table 9). The certificate of Proposition 2 was at most  $3.3 \times 10^{-4}$  nats over all 100 MSX fits.

Table 1: Synthetic recovery: held-out log-likelihood deficit to the true model ( $10^{-3}$  nats per event, lower is better; mean  $\pm$  s.d. over five seeds). Bold: best; underlined: second best; shaded: our estimators.

|                             | S1 exp                 | S2 power law           | S3 hump                | S4 multiscale          | S5 network             |
|-----------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|
| <i>Classical estimators</i> |                        |                        |                        |                        |                        |
| Exp. kernel MLE             | <b>0.05</b> $\pm 0.05$ | 26.05 $\pm 0.50$       | 101.47 $\pm 1.81$      | 711.28 $\pm 53.09$     | 426.80 $\pm 368.57$    |
| Sum-of-exp. MLE             | 0.92 $\pm 0.13$        | <b>0.18</b> $\pm 0.12$ | 11.51 $\pm 0.46$       | —                      | —                      |
| ADM4                        | <u>0.05</u> $\pm 0.05$ | 26.05 $\pm 0.50$       | 101.47 $\pm 1.81$      | 3.13 $\pm 0.25$        | 26.51 $\pm 0.29$       |
| Hawkes EM (nonparam.)       | 1.02 $\pm 0.07$        | 1.73 $\pm 0.21$        | 2.24 $\pm 0.12$        | 4.55 $\pm 0.29$        | 5.23 $\pm 0.25$        |
| Conditional law             | 2.84 $\pm 1.03$        | 4.91 $\pm 0.59$        | 10.66 $\pm 2.13$       | 114.94 $\pm 1.20$      | 78.64 $\pm 4.79$       |
| <i>Ours</i>                 |                        |                        |                        |                        |                        |
| MSX ( $R = 2$ )             | 0.23 $\pm 0.12$        | 0.32 $\pm 0.12$        | 5.90 $\pm 0.36$        | 0.54 $\pm 0.12$        | 1.19 $\pm 0.07$        |
| MSX ( $R = 4$ )             | 0.28 $\pm 0.15$        | 0.45 $\pm 0.10$        | <u>0.31</u> $\pm 0.15$ | 0.70 $\pm 0.11$        | 1.63 $\pm 0.09$        |
| MSX, exponential only       | 0.91 $\pm 0.13$        | <u>0.19</u> $\pm 0.11$ | 11.51 $\pm 0.46$       | <u>0.32</u> $\pm 0.08$ | <u>0.82</u> $\pm 0.08$ |
| MSX-auto                    | 0.24 $\pm 0.12$        | 0.33 $\pm 0.13$        | <b>0.30</b> $\pm 0.13$ | <b>0.32</b> $\pm 0.08$ | <b>0.74</b> $\pm 0.10$ |

## 6.2 Public temporal point process benchmarks

We use the EasyTPP splits of Amazon, Retweet, Taxi, Taobao and StackOverflow and the protocol of Chang et al. (2025): log-likelihood per scored event with the first event of each sequence as conditioning and model selection on validation likelihood. Published numbers are quoted from Chang et al. (2025); our re-run of S2P2 on Taxi gives 0.524 against the published 0.522, validating the harness. EPT-X reports the exact likelihood.

On Retweet, the largest dataset (2.2 million training events), EPT-X improves on the best published model by more than 0.6 nats per event, with a 25-epoch budget imposed by compute and the validation likelihood still rising (Table 2). It is statistically tied with the best published model on Taobao and StackOverflow and close on Taxi, where the time term equals S2P2’s on our re-run (0.733 vs. 0.735 nats per event) and the difference lies entirely in the mark term. On Amazon the gaps concentrate in a narrow band (median 0.73, 99th percentile 0.80); Erlang orders up to 16 cannot represent the mode, and atoms of order up to 1024 on a finer grid raise EPT-X by more than 1.1 nats per event, into second place. Here, in contrast to Taxi, our mark term is better than S2P2’s ( $-1.87$  vs.  $-1.90$  on the re-run) and the remaining gap is in the time term. The linear MSX model is close to the neural models on Taobao and Retweet and far behind on Amazon and Taxi, whose dynamics are not self-exciting.

Table 2: Public benchmarks: test log-likelihood per event (higher is better). Published numbers are from Chang et al. (2025) (five seeds each); our rows give mean  $\pm$  s.d. over seeds (EPT-X: Amazon three; Retweet three; Taxi five; Taobao three; StackOverflow three; other rows one seed), with the EPT-X configuration selected per dataset on validation. Bold: best; underlined: second best; shaded: our models.

|                                                    | Amazon                   | Retweet                   | Taxi              | Taobao                   | StackOverflow             |
|----------------------------------------------------|--------------------------|---------------------------|-------------------|--------------------------|---------------------------|
| <i>Neural TPPs, published (Chang et al., 2025)</i> |                          |                           |                   |                          |                           |
| RMTTPP                                             | −2.136                   | −7.098                    | 0.346             | 1.003                    | −2.480                    |
| SAHP                                               | −2.074                   | −6.708                    | 0.298             | 1.168                    | −2.341                    |
| THP                                                | −2.096                   | −6.659                    | 0.372             | 0.790                    | −2.338                    |
| IntensityFree                                      | 0.496                    | −10.344                   | 0.453             | <b>1.318</b>             | −2.233                    |
| MHP                                                | −2.091                   | −6.564                    | 0.370             | 0.636                    | −2.346                    |
| NHP                                                | 0.129                    | <u>−6.348</u>             | <u>0.514</u>      | 1.157                    | −2.241                    |
| AttNHP                                             | 0.484                    | −6.499                    | 0.493             | 1.259                    | −2.194                    |
| S2P2                                               | <b>0.781</b>             | −6.365                    | <b>0.522</b>      | 1.304                    | <b>−2.163</b>             |
| <i>Classical Hawkes, our runs</i>                  |                          |                           |                   |                          |                           |
| ADM4                                               | −2.260                   | −6.832                    | −0.315            | 1.184                    | −2.473                    |
| Conditional law                                    | −                        | −8.317                    | −2.922            | −63.919                  | −16.140                   |
| Hawkes EM (nonparam.)                              | −1.002                   | −6.655                    | −0.308            | 1.269                    | −2.370                    |
| <i>Ours</i>                                        |                          |                           |                   |                          |                           |
| MSX ( $R = 2$ )                                    | −1.943                   | −6.569                    | −0.214            | 1.299                    | −2.292                    |
| MSX-auto                                           | −1.927                   | −6.568                    | −0.193            | 1.305                    | −2.292                    |
| EPT                                                | −1.203                   | −                         | 0.437             | 1.311                    | −2.186                    |
| EPT-X                                              | <u>0.630</u> $\pm 0.003$ | <b>−5.716</b> $\pm 0.024$ | 0.503 $\pm 0.004$ | <u>1.316</u> $\pm 0.005$ | <u>−2.166</u> $\pm 0.001$ |

### 6.3 Limit-order-book event streams

We use the LOBSTER samples of AAPL, AMZN, GOOG, INTC and MSFT (21 June 2012), six event types, a chronological 60/20/20 split and sequences of 128 events. Tied timestamps, which EasyTPP rejects, are displaced by  $U(0, 100\text{ns})$  in the files that every model is trained and scored on. Neural baselines run in EasyTPP with the S2P2 authors’ configurations (batch 32–64 instead of 256 on an 8 GB GPU); each is trained with a 100-epoch budget, the strongest additionally with longer budgets, and reported at the budget with the best validation likelihood, so longer training can only help them.

EPT-X is the best model on every stock (Table 3), ahead of MSX-auto, EPT and all seven neural baselines, by 0.23–0.45 nats per event over the best neural baseline. One baseline needs a correction. EasyTPP’s IntensityFree clamps every inter-event time at  $10^{-5}\text{s}$  before evaluating its log-normal-mixture density, so all shorter gaps are scored as one common value, on which the mixture can place an arbitrarily narrow component. This is not the likelihood of the observed gaps, and it matters where short gaps are common: 29% and 32% of test gaps on INTC and MSFT lie below  $10^{-5}\text{s}$ , against 10–14% on the other stocks. As shipped, IntensityFree reports the highest numbers on INTC and MSFT; re-trained and scored on the unclamped gaps (same 1500-epoch budget) it loses 1.1 and 0.7 nats per event and falls below EPT-X, and on INTC below MSX. Consistently, a denser MSX dictionary and sharper EPT-X atoms change the INTC validation likelihood by at most about 0.05 nats per event, an order of magnitude less than the effect of the clamp. We report both IntensityFree variants and exclude the clamped one from the ranking; the patch is part of the supplementary code.

Table 3: LOBSTER order books: test log-likelihood per event (higher is better; mean  $\pm$  s.d. over seeds where several were run). Bold: best; underlined: second best; shaded: our models. \*As shipped, IntensityFree clamps gaps at  $10^{-5}$  s before scoring, which is not the likelihood of the observed data; it is shown in grey and excluded from the ranking, and the IntensityFree row is re-trained and scored on the unclamped gaps.

|                                        | AAPL                     | AMZN                     | GOOG                     | INTC                     | MSFT                     |
|----------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| <i>Classical Hawkes</i>                |                          |                          |                          |                          |                          |
| Exp. kernel MLE                        | 1.961                    | –                        | –0.311                   | –                        | –                        |
| Hawkes EM (nonparam.)                  | 3.370                    | 2.243                    | 2.851                    | 5.771                    | 6.258                    |
| ADM4                                   | 1.961                    | 1.067                    | 1.306                    | 4.787                    | 5.211                    |
| Conditional law                        | –1.018                   | –17.814                  | –32.784                  | 0.379                    | –1.270                   |
| <i>Neural TPPs (EasyTPP, our runs)</i> |                          |                          |                          |                          |                          |
| RMTTP                                  | 0.243                    | –0.686                   | –0.948                   | 2.882                    | 3.285                    |
| THP                                    | –0.081                   | –0.948                   | –1.075                   | 1.714                    | 1.855                    |
| SAHP                                   | 1.777                    | 0.462                    | –0.428                   | 3.915                    | 4.557                    |
| NHP                                    | 1.374                    | 0.842                    | 0.846                    | 3.874                    | 4.183                    |
| AttNHP                                 | 1.220                    | –0.168                   | 0.004                    | 3.497                    | 3.771                    |
| S2P2                                   | 2.315                    | 0.717                    | 0.990                    | 4.949                    | 5.416                    |
| IntensityFree                          | 3.108                    | 2.103                    | 2.726                    | 5.713                    | 6.377                    |
| IntensityFree* (as shipped)            | 3.260                    | 1.921                    | 2.679                    | 6.790                    | 7.070                    |
| <i>Ours</i>                            |                          |                          |                          |                          |                          |
| MSX ( $R = 2$ )                        | 3.326                    | 2.249                    | 2.870                    | 5.812                    | 6.307                    |
| MSX-auto                               | 3.361                    | <u>2.281</u>             | <u>2.883</u>             | 5.814                    | 6.310                    |
| EPT                                    | <u>3.380</u> $\pm 0.008$ | 2.229 $\pm 0.003$        | 2.806 $\pm 0.005$        | <u>6.024</u> $\pm 0.007$ | <u>6.530</u> $\pm 0.005$ |
| EPT-X                                  | <b>3.534</b> $\pm 0.016$ | <b>2.332</b> $\pm 0.005$ | <b>2.992</b> $\pm 0.012$ | <b>6.163</b> $\pm 0.002$ | <b>6.666</b> $\pm 0.001$ |

## 6.4 Ablations

Table 4 isolates the components of EPT-X on validation data. The renewal channel improves EPT on every dataset. Sharp atoms help where the gap law has sharp modes—by 1.1 nats per event on Amazon, where a finer grid helps further, and by 0.05 on INTC—and are neutral or slightly worse elsewhere. Neither a deeper (three residual layers with LayerNorm and dropout) nor a wider (128 units) encoder improves any dataset, nor does S2P2’s larger batch: S2P2’s remaining edge on Taxi is not a matter of encoder capacity or optimisation protocol. Without the explicit Hawkes backbone, EPT reached 2.84 nats per event on LOBSTER AAPL, below the classical MSX estimator.

Table 4: EPT-X ablations: validation log-likelihood per event (higher is better). EPT and EPT-X rows are the final runs (mean  $\pm$  s.d. over seeds); variants are single-seed selection runs with a 100-epoch budget. Bold: best; underlined: second best.

|                                            | Taxi                     | Taobao                   | StackOverflow             | Amazon                    | INTC                     |
|--------------------------------------------|--------------------------|--------------------------|---------------------------|---------------------------|--------------------------|
| <i>Validation log-likelihood per event</i> |                          |                          |                           |                           |                          |
| EPT (no renewal channel)                   | 0.392                    | <u>1.200</u>             | –2.186                    | –1.193                    | 5.360 $\pm 0.007$        |
| EPT-X                                      | <b>0.466</b> $\pm 0.004$ | <b>1.205</b> $\pm 0.002$ | <b>–2.161</b> $\pm 0.003$ | <b>–0.487</b> $\pm 0.000$ | <b>5.486</b> $\pm 0.025$ |
| + sharp atoms (orders to 1024)             | <u>0.457</u>             | 1.194                    | <u>–2.173</u>             | <u>0.604</u>              | <b>5.516</b>             |
| + sharp atoms, 48 scales                   | 0.455                    | 1.177                    | –2.182                    | <b>0.643</b>              | <u>5.511</u>             |
| + deep encoder (3 layers)                  | 0.396                    | 1.142                    | –2.264                    | –0.507                    | 5.461                    |
| + sharp atoms + deep encoder               | 0.398                    | 1.152                    | –2.248                    | 0.579                     | 5.496                    |
| + wide encoder (128 units)                 | 0.431                    | 1.188                    | –2.195                    | –0.489                    | 5.459                    |
| batch 256 (S2P2 setting)                   | 0.440                    | 1.189                    | –2.442                    | –0.518                    | 5.435                    |

## 7 Application: how fast do markets absorb macroeconomic news?

**Data.** The calendar covers January 2022–August 2026: 670 scheduled releases of CPI, nonfarm payrolls, PPI, retail sales, the GDP advance estimate and initial jobless claims (dated by their initial ALFRED vintage), FOMC statements and press conferences, ECB decisions and press conferences, and Bank of Japan decisions. Surprises are standardised, sign-split differences between the first print and a real-time expectation: the Cleveland Fed inflation nowcast (CPI), Atlanta Fed GDPNow (GDP) and real-time statistical expectations from earlier first prints otherwise. Prices are millisecond quotes for EUR/USD, USD/JPY, gold and an S&P 500 CFD, and Binance futures trades for bitcoin and ether, converted to up and down  $\delta$ -crossing events ( $\delta = 0.25\text{--}3$  basis points per asset, chosen so that the median rate is about 0.2 events per second). Windows span 30 minutes before to 60 minutes after each release cluster, with 30 minutes of burn-in. Each window is matched with a placebo window at the same clock times on the same weekday one to six weeks away, with no modelled release and none of sixteen second-tier releases within three hours; placebo windows carry their own per-type pseudo-releases, so the test is part of the likelihood. The pooled model has 732 windows (430 release, 302 placebo), 7.3 million scored events and twelve dimensions, and is certified optimal to within 0.0054 nats per dimension.

**Endogeneity.** The spectral radius of the branching matrix is  $\rho(G) = 0.87$ : roughly seven in every eight price events are echoes of earlier ones. Fitted year by year,  $\rho(G)$  declines from 2022 to 2026 (0.91, 0.89, 0.85, 0.86, 0.85), and the slowest echo mode relaxes in about 12 minutes.

**Falsification.** Nonnegative news kernels have a positive expected mass even under the null (boundary bias), so a release-free placebo is essential. With an exposure-weighted  $\ell_1$  level chosen by held-out likelihood, 5 of the 11 release types exceed their placebo by at least 50% (Figure 1): CPI (8.5 $\times$ ), payrolls (6.7 $\times$ ), PPI (4.0 $\times$ ), the FOMC statement (3.9 $\times$ ) and retail sales (1.9 $\times$ ). Jobless claims, ECB and BoJ decisions and the press conferences have no detectable direct effect beyond their placebos. Two earlier specifications failed this test and are reported in Appendix B.

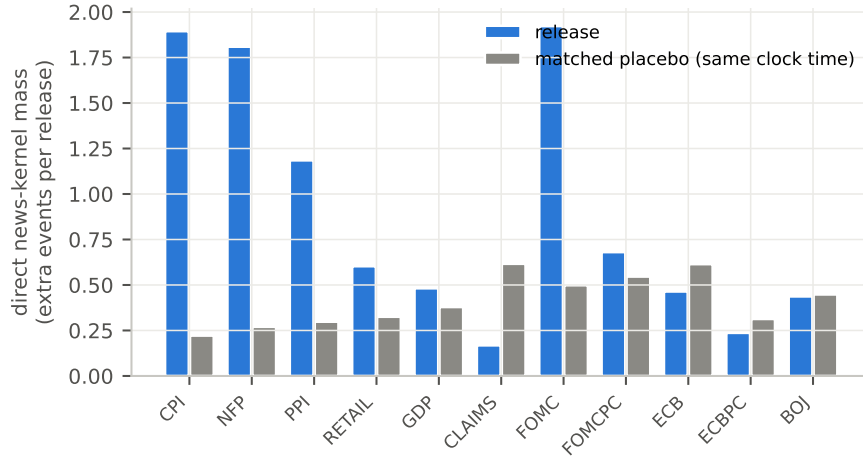


Figure 1: Placebo test: direct news-kernel mass (expected extra events per release, all assets) for each release type and for its matched placebo on release-free days.

**Echo-corrected absorption.** For the releases that pass, the direct news kernel is absorbed almost instantly (median  $t_{50}$  of 1.1 s), but echoes stretch the response: half of the extra activity arrives within 16–137 s and ninety percent within 5–11 minutes (Figure 2, Table 5). The “news half-life”  $\ln 2/\beta$  therefore understates absorption time by one to two orders of magnitude. Because signed surprises enter the news kernel separately by sign, the model also gives price-discovery curves (Figure 3, Table 6); hotter inflation lowers bitcoin and ether, cooler inflation lifts them.

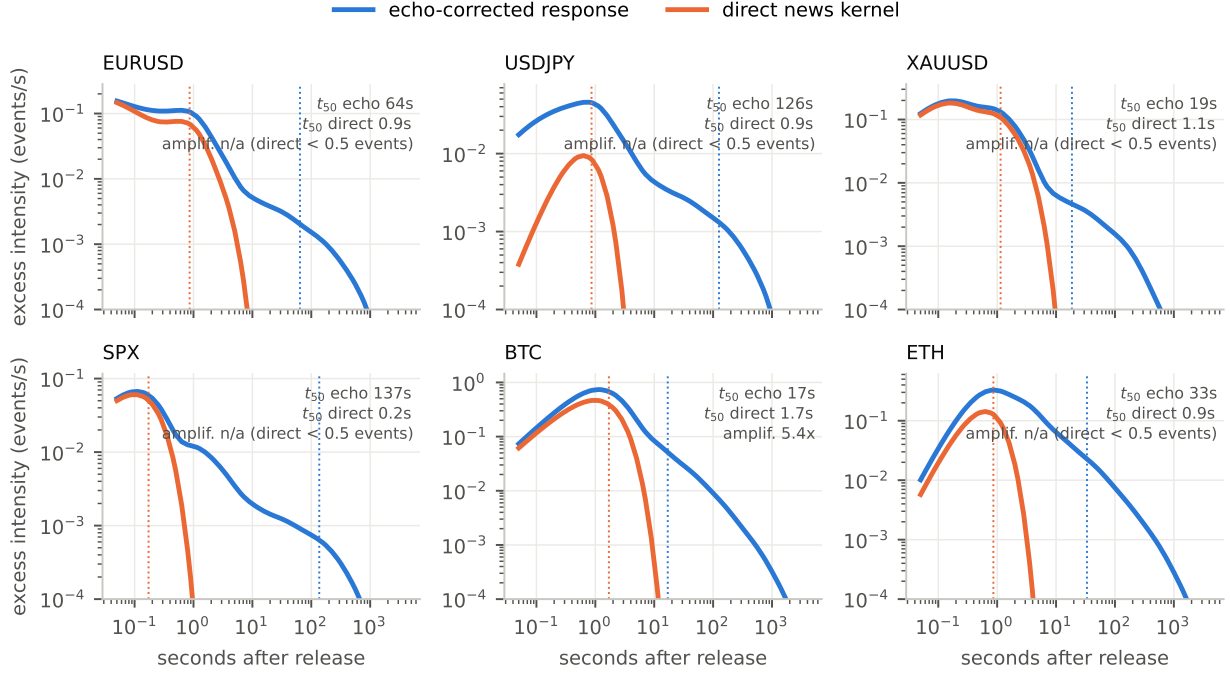


Figure 2: Echo-corrected response to a CPI release (blue) versus the direct news kernel (orange), per asset; dotted lines mark  $t_{50}$ .

Table 5: Echo-corrected absorption by release and asset:  $t_{50}$  in seconds /  $t_{90}$  in minutes. \*Release types that do not exceed their placebo by 50%.

| Release           | EURUSD     | USDJPY     | Gold     | S&P 500    | BTC      | ETH       |
|-------------------|------------|------------|----------|------------|----------|-----------|
| CPI               | 64 / 9.1   | 126 / 11.0 | 19 / 4.8 | 137 / 11.0 | 17 / 6.1 | 33 / 7.8  |
| NFP               | 94 / 10.1  | 109 / 10.7 | 16 / 5.2 | 80 / 8.8   | 27 / 7.5 | 24 / 7.3  |
| PPI               | 120 / 10.8 | 94 / 10.3  | 40 / 6.4 | 79 / 8.6   | 23 / 7.2 | 39 / 8.7  |
| FOMC              | 89 / 10.1  | 76 / 9.8   | 82 / 9.0 | 77 / 8.9   | 41 / 9.2 | 36 / 9.1  |
| Retail            | 59 / 9.1   | 77 / 9.8   | 87 / 9.2 | 114 / 10.4 | 26 / 7.6 | 26 / 7.8  |
| GDP*              | 95 / 10.0  | 81 / 9.6   | 26 / 5.1 | 145 / 11.2 | 27 / 7.3 | 27 / 7.5  |
| FOMC press conf.* | 88 / 10.0  | 111 / 10.9 | 60 / 8.1 | 59 / 8.0   | 45 / 9.5 | 52 / 10.0 |
| Claims*           | 124 / 11.0 | 56 / 9.1   | 87 / 9.3 | 81 / 9.1   | 36 / 9.0 | 40 / 9.4  |
| ECB*              | 61 / 9.2   | 73 / 9.6   | 88 / 9.0 | 123 / 10.8 | 37 / 9.2 | 42 / 9.6  |
| ECB press conf.*  | 137 / 11.3 | 90 / 10.2  | 38 / 6.5 | 63 / 8.2   | 25 / 7.3 | 32 / 8.0  |
| BoJ*              | 93 / 10.2  | 80 / 9.9   | 45 / 7.3 | 73 / 8.8   | 35 / 8.5 | 37 / 8.8  |

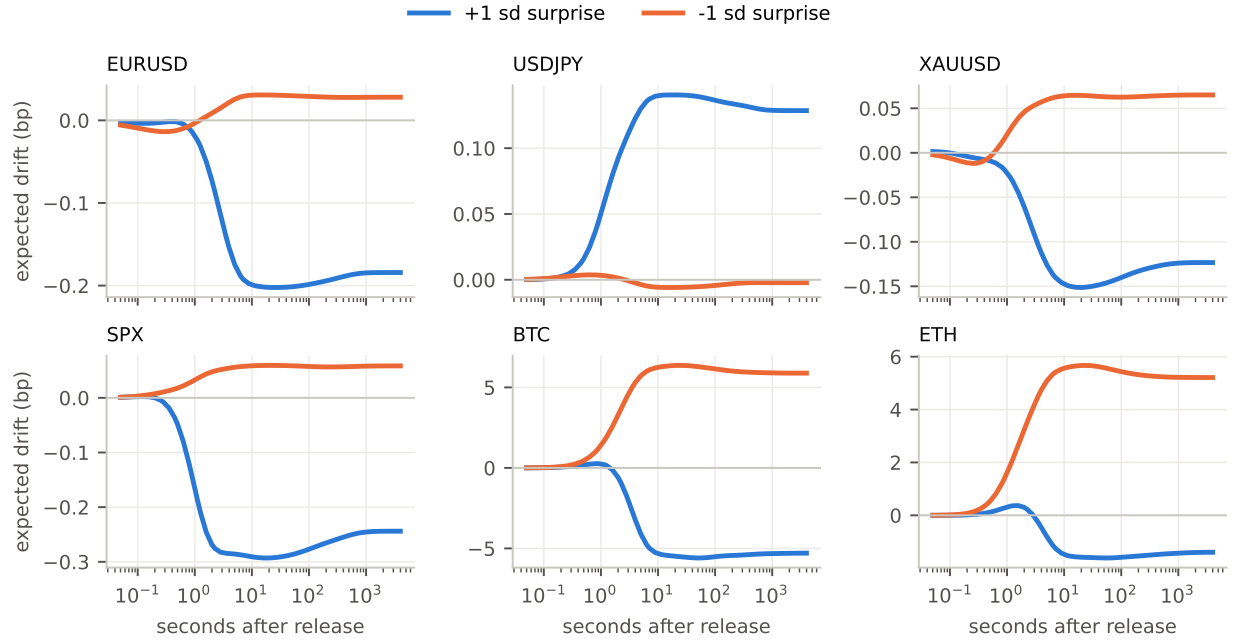
Figure 3: Expected price-discovery curves after a  $\pm 1$  standard-deviation CPI surprise.

Table 6: Expected price drift (basis points, end of window) after a +1 standard-deviation surprise. \*As in Table 5.

| Release           | EURUSD | USDJPY | Gold | S&P 500 | BTC  | ETH  |
|-------------------|--------|--------|------|---------|------|------|
| CPI               | -0.2   | +0.1   | -0.1 | -0.2    | -5.3 | -1.4 |
| NFP               | -0.1   | +0.1   | +0.1 | +0.2    | -0.3 | -1.5 |
| PPI               | 0.0    | +0.1   | -0.2 | -0.3    | -2.4 | -2.1 |
| FOMC              | 0.0    | -0.1   | +0.1 | +0.1    | -0.3 | +0.2 |
| Retail            | +0.1   | -0.1   | +0.2 | 0.0     | -1.3 | -0.5 |
| GDP*              | +0.1   | -0.1   | -0.1 | 0.0     | +0.9 | +0.2 |
| FOMC press conf.* | 0.0    | 0.0    | 0.0  | +0.1    | 0.0  | +0.1 |
| Claims*           | 0.0    | 0.0    | 0.0  | 0.0     | +0.4 | +0.6 |
| ECB*              | 0.0    | 0.0    | 0.0  | 0.0     | -0.1 | -0.1 |
| ECB press conf.*  | 0.0    | 0.0    | 0.0  | 0.0     | -0.3 | -0.3 |
| BoJ*              | 0.0    | 0.0    | 0.0  | 0.0     | -0.1 | +0.3 |

**Goodness of fit.** Time-rescaled residuals (Brown et al., 2002) have Kolmogorov–Smirnov statistics of at most 0.044 against  $\text{Exp}(1)$ ; the per-window excess-dispersion test rejects in 26–55% of windows, compared with 42% (power-law kernel) and 93% (double exponential) reported for FX data by Rambaldi et al. (2015).

**Forecasting the post-release burst.** On the 163 releases of 2025–2026, an MSX model fitted on 2022–2024 forecasts the number of  $\delta$ -crossings in the first 1, 5 and 15 minutes in closed form (Table 7). It improves on same-type climatology at every horizon but is beaten by the event-study OLS and HAR regressions, which are fitted directly to this target; we report this negative result as is. For the direction of the move the structural model is informative: the correlation between its forecast and the realised one-minute drift is 0.17, against 0.06 for the best baseline.

Table 7: Post-release activity forecasts on the 163 test releases (mean over assets; lower is better). Bold: best; underlined: second best; shaded: our model.

| Method          | RMSE of $\log(1 + N)$ |              |              | QLIKE of realised variance |              |              |
|-----------------|-----------------------|--------------|--------------|----------------------------|--------------|--------------|
|                 | 1 min                 | 5 min        | 15 min       | 1 min                      | 5 min        | 15 min       |
| Climatology     | 1.440                 | 1.308        | 1.199        | <b>1.263</b>               | 0.916        | 0.701        |
| Event-study OLS | <b>1.140</b>          | <b>0.925</b> | <b>0.776</b> | 2.552                      | <b>0.580</b> | <b>0.372</b> |
| HAR             | <u>1.217</u>          | <u>0.985</u> | <u>0.820</u> | <u>1.467</u>               | <u>0.762</u> | <u>0.468</u> |
| MSX closed form | 1.322                 | 1.049        | 0.955        | 2.913                      | 1.036        | 0.518        |

## 8 Discussion and limitations

Making every kernel a nonnegative combination of phase-type densities turns Hawkes estimation into a certified convex problem and turns responses to shocks into matrix exponentials; adding a neural history encoder and a defective hyper-Erlang renewal channel to the same state space gives a model whose likelihood stays exact and which is competitive with, and on order books and Retweet better than, the strongest neural temporal point processes. Three lessons stand out. First, absorption speed is not a property of the news kernel: echoes lengthen it by one to two orders of magnitude, and the correction is available in closed form. Second, falsification matters: two of our specifications produced plausible-looking news effects that the placebo test exposed as artefacts. Third, benchmark comparisons must check that every model scores the same observed data—a clamp inside one implementation was enough to reverse a ranking.

**Limitations.** Expectations for payrolls, claims, PPI and retail sales are statistical rather than survey-based, and the FX feed has a gap in the first half of 2023. The order-book evaluation uses the public LOBSTER samples (one trading day per stock). Neural baselines on order books were trained with reduced batch sizes on an 8 GB GPU, and on the public benchmarks we compare against published numbers, validated by one re-run. On Retweet the EPT-X budget was 25 epochs with the validation likelihood still rising, so the reported margin is if anything conservative. EPT-X does not reach the best published likelihood on Taxi, where the gap lies in mark prediction, or on Amazon, where it lies in the timing density; and the MSX forecasts of post-release activity are beaten by regressions fitted to that target.

### Broader Impact Statement

This work models market event streams with public or licensable data and makes no use of personal data. Better models of how quickly prices absorb public information can inform market monitoring and the timing of disclosures, and exact likelihoods make benchmark comparisons more reliable. Faster or more accurate event forecasts could also be used in trading; the methods described are standard statistical tools and do not by themselves confer an informational advantage. We release code and patches so that the reported numbers, including the corrected baseline, can be reproduced.

## References

- Massil Achab, Emmanuel Bacry, Stéphane Gaïffas, Iacopo Mastromatteo, and Jean-François Muzy. Uncovering causality from multivariate Hawkes integrated cumulants. *Journal of Machine Learning Research*, 18(192):1–28, 2018.
- Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Clara Vega. Micro effects of macro announcements: Real-time price discovery in foreign exchange. *American Economic Review*, 93(1):38–62, 2003.
- Emmanuel Bacry, Iacopo Mastromatteo, and Jean-François Muzy. Hawkes processes in finance. *Market Microstructure and Liquidity*, 1(01):1550005, 2015.
- Emmanuel Bacry, Thibault Jaisson, and Jean-François Muzy. Estimation of slowly decreasing Hawkes kernels: application to high-frequency order book dynamics. *Quantitative Finance*, 16(8):1179–1201, 2016.
- Tanguy Bosser and Souhaib Ben Taieb. On the predictive accuracy of neural temporal point process models for continuous-time event data. *TMLR*, 2023.
- Alex Boyd, Andrew Warrington, Taha Kass-Hout, Parminder Bhatia, and Danica Xiao. Hyper Hawkes processes: Interpretable models of marked temporal point processes. *arXiv:2511.01096*, 2025.
- Emery N. Brown, Riccardo Barbieri, Valérie Ventura, Robert E. Kass, and Loren M. Frank. The time-rescaling theorem and its application to neural spike train data analysis. *Neural Computation*, 14(2):325–346, 2002.
- Yuxin Chang et al. Deep continuous-time state-space models for marked event sequences. *arXiv:2412.19634*, 2025.
- Nan Du, Hanjun Dai, Rakshit Trivedi, Utkarsh Upadhyay, Manuel Gomez-Rodriguez, and Le Song. Recurrent marked temporal point processes: Embedding event history to vector. In *KDD*, 2016.
- Vladimir Filimonov and Didier Sornette. Quantifying reflexivity in financial markets: Toward a prediction of flash crashes. *Physical Review E*, 85:056108, 2012.
- Vladimir Filimonov and Didier Sornette. Apparent criticality and calibration issues in the Hawkes self-excited point process model: application to high-frequency financial data. *Quantitative Finance*, 15(8):1293–1314, 2015.
- Anningzhe Gao et al. Mamba Hawkes process. *arXiv:2407.05302*, 2024.
- Stephen J. Hardiman, Nicolas Bercot, and Jean-Philippe Bouchaud. Critical reflexivity in financial markets: a Hawkes process analysis. *The European Physical Journal B*, 86:442, 2013.
- Alan G. Hawkes. Spectra of some self-exciting and mutually exciting point processes. *Biometrika*, 58(1):83–90, 1971.
- Matthias Kirchner. An estimation procedure for the Hawkes process. *Quantitative Finance*, 17(4):571–595, 2017.
- Simon C. K. Lee and X. Sheldon Lin. Modeling and evaluating insurance losses via mixtures of Erlang distributions. *North American Actuarial Journal*, 14(1):107–130, 2010.
- Erik Lewis and George Mohler. A nonparametric EM algorithm for multiscale Hawkes processes. *Journal of Nonparametric Statistics*, 1(1):1–20, 2011.
- Hongyuan Mei and Jason Eisner. The neural Hawkes process: A neurally self-modulating multivariate point process. In *NeurIPS*, 2017.
- Takahiro Omi, Yoshito Hirata, and Kazuyuki Aihara. Hawkes process model with a time-dependent background rate and its application to high-frequency financial data. *Physical Review E*, 96:012303, 2017.

- Marcello Rambaldi, Paris Pennesi, and Fabrizio Lillo. Modeling foreign exchange market activity around macroeconomic news: A Hawkes-process approach. *Physical Review E*, 91(1):012819, 2015.
- Oleksandr Shchur, Marin Bilos, and Stephan Günnemann. Intensity-free learning of temporal point processes. In *ICLR*, 2020.
- Henk C. Tijms. *Stochastic Models: An Algorithmic Approach*. Wiley, 1994.
- Ravi Varadhan and Christophe Roland. Simple and globally convergent methods for accelerating the convergence of any EM algorithm. *Scandinavian Journal of Statistics*, 35(2):335–353, 2008.
- Richard S. Varga. *Matrix Iterative Analysis*. Springer, 2000.
- Siqiao Xue et al. EasyTPP: Towards open benchmarking temporal point processes. In *ICLR*, 2024.
- Chenghao Yang, Hongyuan Mei, and Jason Eisner. Transformer embeddings of irregularly spaced events and their participants. In *ICLR*, 2022.
- Qiang Zhang, Aldo Lipani, Omer Kirnap, and Emine Yilmaz. Self-attentive Hawkes process. In *ICML*, 2020.
- Ke Zhou, Hongyuan Zha, and Le Song. Learning social infectivity in sparse low-rank networks using multi-dimensional Hawkes processes. In *AISTATS*, 2013.
- Simiao Zuo, Haoming Jiang, Zichong Li, Tuo Zhao, and Hongyuan Zha. Transformer Hawkes process. In *ICML*, 2020.

## A Proofs

*Proof of Proposition 2.* Fenchel’s inequality  $\log u \leq vu - 1 - \log v$  for  $u, v > 0$ , applied to  $u = \lambda_n$  with  $v = v_n$ , gives  $\sum_n \log \lambda_n \leq \theta^\top X^\top \mathbf{v} - N - \sum_n \log v_n$ . If  $X^\top \mathbf{v} \leq \Phi + \gamma$  then  $\theta^\top X^\top \mathbf{v} \leq \theta^\top (\Phi + \gamma)$  for every  $\theta \geq 0$ , hence  $\ell(\theta) \leq D(\mathbf{v}) := -N - \sum_n \log v_n$  for every feasible  $\theta$ , including the maximiser. The choice  $v_n = 1/(s\lambda_n)$  with  $s$  as stated is dual feasible and gives  $D(\mathbf{v}) - \ell(\theta) = N \log s + \theta^\top (\Phi + \gamma) - N$ . At a KKT point  $s = 1$  and  $\theta^\top (\Phi + \gamma) = \sum_n \theta^\top x_n / \lambda_n = N$ , so both terms vanish.  $\square$

## B Specification history of the application

We report the specifications that failed the built-in falsification test, because the final specification was chosen by them rather than by the results we hoped for.

**Version 1.** News kernels with time scales up to 30 minutes and Erlang order 3, a constant baseline per window, time-of-day hats in UTC, and tied events (a trade sweeping several  $\delta$  levels produced several events with the same timestamp). The single placebo type absorbed 169 extra events per pseudo-release, comparable to real releases, and crypto residuals had Kolmogorov–Smirnov statistics up to 0.58. News kernels longer than the post-release window acted as trend absorbers, and ties violated orderliness.

**Version 2.** One event per timestamp with a size mark, news kernels restricted to 0.1–300 s, a per-window linear baseline and New-York-time seasonality. The placebo mass fell to 18.6, but remained significant; placebo mass appeared at every release slot, which ruled out unmodelled 08:30 releases as the whole explanation.

**Version 3 (final).** Placebo days also avoid sixteen second-tier US releases, and nonnegative news kernels carry an exposure-weighted  $\ell_1$  penalty chosen by held-out profile likelihood (Table 8); without it, release and placebo masses are equally inflated by boundary bias. The held-out likelihood peaks at  $\ell_1 = 0.8$ , where release and placebo masses separate.

Table 8: Selection of the news-kernel penalty on held-out 2024 windows (profile log-likelihood relative to the best level).

| $\ell_1$ level | held-out LL (rel.) | mean release mass | mean placebo mass |
|----------------|--------------------|-------------------|-------------------|
| 0.2            | -93.7              | 10.79             | 6.23              |
| 0.4            | -27.1              | 5.20              | 2.89              |
| 0.8            | <b>0.0</b>         | 1.81              | 1.03              |
| 1.6            | -20.5              | 0.44              | 0.37              |

## C Synthetic recovery: all metrics

Table 9: Synthetic recovery, all metrics (mean  $\pm$  s.d. over five seeds): relative branching-matrix error, spectral-radius error, kernel  $L^1$  error, held-out deficit dLL ( $10^{-3}$  nats/event) and edge AUC.

| Scenario      | Method                        | $G$ err.          | $\rho$ err.       | kernel $L^1$ | dLL ( $10^{-3}$ )                 | AUC   |
|---------------|-------------------------------|-------------------|-------------------|--------------|-----------------------------------|-------|
| S1-exp        | MSX-auto (ours)               | $0.054 \pm 0.032$ | $0.020 \pm 0.018$ | 0.093        | $0.24 \pm 0.12$                   | –     |
|               | MSX (ours)                    | $0.056 \pm 0.032$ | $0.019 \pm 0.018$ | 0.092        | $0.23 \pm 0.12$                   | –     |
|               | MSX-R4 (ours, fixed Erlang-4) | $0.056 \pm 0.035$ | $0.023 \pm 0.018$ | 0.105        | $0.28 \pm 0.15$                   | –     |
|               | MSX-exp (ablation: no Erlang) | $0.055 \pm 0.025$ | $0.026 \pm 0.015$ | 0.157        | $0.91 \pm 0.13$                   | –     |
|               | ExpKern(MLE, best decay)      | $0.018 \pm 0.007$ | $0.005 \pm 0.003$ | 0.018        | <b><math>0.05 \pm 0.05</math></b> | –     |
|               | SumExpKern(MLE, AGD)          | $0.065 \pm 0.022$ | $0.030 \pm 0.013$ | 0.165        | $0.92 \pm 0.13$                   | –     |
|               | ADM4                          | $0.018 \pm 0.007$ | $0.005 \pm 0.003$ | 0.018        | $0.05 \pm 0.05$                   | –     |
|               | EM(nonparam)                  | $0.035 \pm 0.015$ | $0.017 \pm 0.007$ | 0.173        | $1.02 \pm 0.07$                   | –     |
|               | NPHC(cumulants)               | $0.247 \pm 0.087$ | $0.032 \pm 0.024$ | –            | –                                 | –     |
|               | ConditionalLaw                | $0.120 \pm 0.054$ | $0.017 \pm 0.016$ | 0.193        | $2.84 \pm 1.03$                   | –     |
| S2-powerlaw   | MSX-auto (ours)               | $0.064 \pm 0.029$ | $0.037 \pm 0.029$ | 0.108        | $0.33 \pm 0.13$                   | –     |
|               | MSX (ours)                    | $0.057 \pm 0.021$ | $0.015 \pm 0.010$ | 0.114        | $0.32 \pm 0.12$                   | –     |
|               | MSX-R4 (ours, fixed Erlang-4) | $0.062 \pm 0.014$ | $0.016 \pm 0.006$ | 0.131        | $0.45 \pm 0.10$                   | –     |
|               | MSX-exp (ablation: no Erlang) | $0.054 \pm 0.022$ | $0.014 \pm 0.010$ | 0.097        | $0.19 \pm 0.11$                   | –     |
|               | ExpKern(MLE, best decay)      | $0.299 \pm 0.011$ | $0.219 \pm 0.007$ | 0.465        | $26.05 \pm 0.50$                  | –     |
|               | SumExpKern(MLE, AGD)          | $0.057 \pm 0.012$ | $0.016 \pm 0.007$ | 0.094        | <b><math>0.18 \pm 0.12</math></b> | –     |
|               | ADM4                          | $0.299 \pm 0.011$ | $0.219 \pm 0.007$ | 0.465        | $26.05 \pm 0.50$                  | –     |
|               | EM(nonparam)                  | $0.057 \pm 0.014$ | $0.037 \pm 0.008$ | 0.188        | $1.73 \pm 0.21$                   | –     |
|               | NPHC(cumulants)               | $0.153 \pm 0.154$ | $0.041 \pm 0.047$ | –            | –                                 | –     |
|               | ConditionalLaw                | $0.102 \pm 0.067$ | $0.029 \pm 0.019$ | 0.291        | $4.91 \pm 0.59$                   | –     |
| S3-hump       | MSX-auto (ours)               | $0.032 \pm 0.016$ | $0.016 \pm 0.007$ | 0.080        | <b><math>0.30 \pm 0.13</math></b> | –     |
|               | MSX (ours)                    | $0.069 \pm 0.009$ | $0.037 \pm 0.004$ | 0.347        | $5.90 \pm 0.36$                   | –     |
|               | MSX-R4 (ours, fixed Erlang-4) | $0.038 \pm 0.010$ | $0.017 \pm 0.005$ | 0.074        | $0.31 \pm 0.15$                   | –     |
|               | MSX-exp (ablation: no Erlang) | $0.034 \pm 0.009$ | $0.015 \pm 0.004$ | 0.416        | $11.51 \pm 0.46$                  | –     |
|               | ExpKern(MLE, best decay)      | $0.166 \pm 0.008$ | $0.053 \pm 0.004$ | 0.801        | $101.47 \pm 1.81$                 | –     |
|               | SumExpKern(MLE, AGD)          | $0.038 \pm 0.011$ | $0.018 \pm 0.005$ | 0.419        | $11.51 \pm 0.46$                  | –     |
|               | ADM4                          | $0.166 \pm 0.008$ | $0.053 \pm 0.004$ | 0.801        | $101.47 \pm 1.81$                 | –     |
|               | EM(nonparam)                  | $0.024 \pm 0.011$ | $0.010 \pm 0.004$ | 0.187        | $2.24 \pm 0.12$                   | –     |
|               | NPHC(cumulants)               | $0.240 \pm 0.096$ | $0.045 \pm 0.048$ | –            | –                                 | –     |
|               | ConditionalLaw                | $0.149 \pm 0.062$ | $0.017 \pm 0.010$ | 0.319        | $10.66 \pm 2.13$                  | –     |
| S4-multiscale | MSX-auto (ours)               | $0.084 \pm 0.014$ | $0.016 \pm 0.006$ | 0.128        | <b><math>0.32 \pm 0.08</math></b> | 1.000 |
|               | MSX (ours)                    | $0.093 \pm 0.015$ | $0.021 \pm 0.005$ | 0.173        | $0.54 \pm 0.12$                   | 1.000 |
|               | MSX-R4 (ours, fixed Erlang-4) | $0.101 \pm 0.019$ | $0.027 \pm 0.007$ | 0.207        | $0.70 \pm 0.11$                   | 1.000 |
|               | MSX-exp (ablation: no Erlang) | $0.084 \pm 0.014$ | $0.016 \pm 0.006$ | 0.128        | <b><math>0.32 \pm 0.08</math></b> | 1.000 |
|               | ExpKern(MLE, best decay)      | $0.173 \pm 0.041$ | $0.095 \pm 0.025$ | 1.635        | $711.28 \pm 53.09$                | 1.000 |
|               | ADM4                          | $0.399 \pm 0.003$ | $0.255 \pm 0.001$ | 0.410        | $3.13 \pm 0.25$                   | 1.000 |
|               | EM(nonparam)                  | $0.153 \pm 0.012$ | $0.045 \pm 0.010$ | 0.358        | $4.55 \pm 0.29$                   | 1.000 |
|               | NPHC(cumulants)               | $1.030 \pm 0.304$ | $0.211 \pm 0.186$ | –            | –                                 | 0.588 |
|               | ConditionalLaw                | $0.464 \pm 0.286$ | $0.139 \pm 0.050$ | 1.059        | $114.94 \pm 1.20$                 | 0.831 |
| S5-network10  | MSX-auto (ours)               | $0.045 \pm 0.004$ | $0.015 \pm 0.005$ | 0.079        | <b><math>0.74 \pm 0.10</math></b> | 1.000 |
|               | MSX (ours)                    | $0.111 \pm 0.013$ | $0.040 \pm 0.013$ | 0.180        | $1.19 \pm 0.07$                   | 1.000 |
|               | MSX-R4 (ours, fixed Erlang-4) | $0.122 \pm 0.008$ | $0.051 \pm 0.013$ | 0.218        | $1.63 \pm 0.09$                   | 1.000 |
|               | MSX-exp (ablation: no Erlang) | $0.099 \pm 0.014$ | $0.030 \pm 0.014$ | 0.145        | $0.82 \pm 0.08$                   | 1.000 |
|               | ExpKern(MLE, best decay)      | $0.337 \pm 0.153$ | $0.103 \pm 0.032$ | 1.302        | $426.80 \pm 368.57$               | 0.977 |
|               | ADM4                          | $0.167 \pm 0.003$ | $0.096 \pm 0.003$ | 0.319        | $26.51 \pm 0.29$                  | 1.000 |
|               | EM(nonparam)                  | $0.085 \pm 0.008$ | $0.045 \pm 0.008$ | 0.280        | $5.23 \pm 0.25$                   | 1.000 |
|               | NPHC(cumulants)               | $1.348 \pm 0.477$ | $0.520 \pm 0.372$ | –            | –                                 | 0.672 |
|               | ConditionalLaw                | $0.461 \pm 0.047$ | $0.037 \pm 0.026$ | 0.641        | $78.64 \pm 4.79$                  | 0.950 |

## D Experimental details

**MSX.** Endogenous dictionary: 10 rates from 5 ms to 600 s, Erlang order 2; exogenous: 8 rates from 0.1 to 300 s, order 3, marks  $(1, z^+, z^-)$ ; anticipation: 4 rates from 10 to 300 s; 96 time-of-day hats; per-window linear baseline. On public and order-book benchmarks MSX uses a shared baseline and a 12-rate grid spanning the 1st percentile to five times the 99.9th percentile of training gaps, and MSX-auto selects the Erlang order in  $\{1, 2, 3, 4\}$  and the  $\ell_1$  level in  $\{0, 0.01, 0.05\}$  on the validation split.

**EPT and EPT-X.** GRU with 64 units; 4 latent channels of 8 learnable rates; Erlang phases selected on validation from  $\{2, 4\}$  (with hidden size  $\{64, 128\}$ ) on Taxi, Taobao and StackOverflow; Hawkes backbone with 10 rates and 2 phases; AdamW with learning rate  $10^{-2}$ , 1% linear warm-up and cosine decay, gradient clipping at 1, batch size 64 (halved automatically on out-of-memory), early stopping on validation likelihood with patience 40 and a 300-epoch budget (25 epochs on Retweet, batch 32). The renewal channel uses 24 log-spaced mean gaps from half the 0.1% to five times the 99.9% quantile of training gaps with orders  $\{1, 4, 16\}$ . Selected on validation, the LOBSTER streams and Amazon add orders up to 1024 and a history-dependent shift, and Amazon uses a finer grid (validation log-likelihood 0.604, 0.643 and 0.653 with 24, 48 and 96 scales). Mark laws are initialised at empirical frequencies.

**Baselines.** Neural baselines run in EasyTPP with the authors’ configuration files; classical Hawkes estimators use the `tick` library (positive-constrained accelerated gradient descent, tolerance  $10^{-10}$ , decay grids matched to MSX). The IntensityFree clamp is made configurable by a two-line patch included in the supplementary code.

**Compute.** All experiments ran on one laptop with an 8 GB GPU and a 28-core CPU; CPU jobs were limited to four threads each. EPT-X runs on the CPU at a speed comparable to the GPU, because its event recursion is sequential.